

AWS EMR Elastic Compute Plane (Spark on YARN)

• r6id.8xlarge Nodes (32 vCPUs, 256 GB RAM, NVMe SSDs) • Zero-Copy Arrow

Spark Master / Driver Node

- Orchestrates Pipeline Controller
- Louvain Modularity Partitioning
- CaaN Super-Node Compression
- Broadcasts Auxiliary Context

Broadcast
Tasks



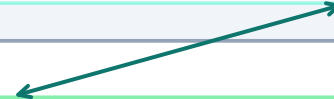
Worker Instances (1 to N Workers)

Spark Executor Container 1

- 28 GB JVM Heap (Spark SQL)
- 12 GB Off-Heap (PyTorch)
- PyArrow Zero-Copy Sharing
- NVMe Storage (/mnt/tmp)

Spark Executor Container 2

- 28 GB JVM Heap (Spark SQL)
- 12 GB Off-Heap (PyTorch)
- PyArrow Zero-Copy Sharing
- NVMe Storage (/mnt/tmp)



Amazon S3 Transactional Storage Plane

Raw Tables

nodes/, edges/, masks/

Community Map

communities/{alg}/

Subgraphs

phase2_nodes, edges

CaaN Macro-Graph

supernodes, minornodes

ACID Commit Log

_delta_log/ (Checkpoints)